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Subject: Digital Image Processing Lab (T7094)

Experiment No. – 3

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Date: 09/07/26

Aim: Write a script to perform point processing for image enhancement (negative, linear stretching, power law transformations)

Software Tools Required: MATLAB

Theory:

Point Processing:

Point processing is the simplest spatial-domain image enhancement technique. It uses a **1×1 neighbourhood**, where each output pixel **s** depends only on the corresponding input pixel **r**. The transformation is given by **$s = T(r)$** . Since pixels are processed independently, it is computationally simple and often used before more advanced enhancement methods.

Negative of an Image:

For an **8-bit grayscale image**, the negative transformation is:

$$s = 255 - r$$

where dark pixels become bright and bright pixels become dark. For **RGB images**, the operation is applied separately to the **Red, Green, and Blue** channels. Negative transformation is useful for highlighting details in dark regions, such as in medical images like mammograms.

Point Processing Techniques:

Common point processing techniques include:

- **Linear transformations:** Identity and Negative transformation.
- **Logarithmic transformation:** Enhances dark regions while compressing bright regions.
- **Power-law (Gamma) transformation:** Brightens or darkens an image depending on the gamma value.
- **Piecewise-linear transformations:** Contrast stretching and gray-level slicing for improving contrast in selected intensity ranges.

Linear Transformation Techniques:

- **Identity Transformation ($s = r$):** Leaves the image unchanged.
- **Negative Transformation ($s = 255 - r$):** Produces the image negative.
- **Contrast Stretching:** Expands a narrow range of gray levels to improve image contrast by increasing the dynamic range.



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Power-Law (Gamma) Transformation:

The transformation is:

$$s = c \times r^\gamma$$

where **c** and **γ (gamma)** are positive constants.

- $\gamma < 1$: Brightens the image by enhancing dark regions.
- $\gamma > 1$: Darkens the image by compressing bright regions.

Gamma transformation is widely used for **gamma correction** to ensure correct image brightness on cameras, monitors, and printers.

Program Code:

```
% Read and convert the grayscale test

image I = imread('cameraman.tif');
I = im2double(I); % Convert to double precision [0,1]

% 1. Negative transformation
neg = imcomplement(I); % Inverts pixel values (s = 1 - r)

% 2. Linear (contrast) stretching
low_in = 0.2;
high_in = 0.8;
low_out = 0;
high_out = 1;
stretch = imadjust(I, [low_in high_in], [low_out high_out]);

% 3. Logarithmic
transformation c = 1 / log(1
+ max(I(:))); logT = c *
log(1 + I);

% 4. Power law (Gamma) transformations
gamma04 = I .^ 0.4;
gamma25 = I .^ 2.5;

figure('Name', 'Intensity Transformations (3x2)', 'Position', [100, 100, 800, 900]);

subplot(3, 2, 1);
imshow(I);
title('Original
Image'); subplot(3,
2, 2); imshow(neg);
title('Negative');
subplot(3, 2, 3);
imshow(stretch);
```



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```
title('Contrast Stretched');  
subplot(3, 2, 4);  
imshow(logT);  
title('Log Transform');  
subplot(3, 2, 5);  
imshow(gamma04);  
title('\gamma = 0.4');  
subplot(3, 2, 6);  
imshow(gamma25);  
title('\gamma = 2.5');
```

Output





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Result Table:

Function	Merits	Demerits	Applications	Alternative
imadjust()	Simple linear contrast stretching	Manual range selection needed	Contrast enhancement	imlincomb()
imcomplement()	Produces image negative easily	May reduce visibility of bright details	Medical imaging (mammograms)	255 - r
im2double()	Converts to double for accurate math	Increases memory usage	Pre-processing for transforms	double()
im2uint8()	Restores 8-bit display range	Clipping of out-of-range values	Display / saving images	uint8()
log()	Expands dark region details	Compresses bright region details	Enhancing dark/underexposed images	log2(), log10()
nthroot() / power law ($r.^{\gamma}$)	Flexible enhancement via gamma	Wrong gamma may over/under enhance	Gamma correction for displays/cameras	imadjust() with gamma param

Conclusion:

The experiment successfully implemented point-based image enhancement techniques in MATLAB, including negative transformation, linear (contrast) stretching, logarithmic transformation and power-law (gamma) transformation. It was observed that the negative transformation is useful for highlighting details embedded in dark regions, contrast stretching improves the visual dynamic range of low-contrast images, and the power-law transformation provides a flexible, tunable mechanism (via the gamma parameter) for either brightening or darkening an image depending on the application and the response characteristics of the display or capture device.